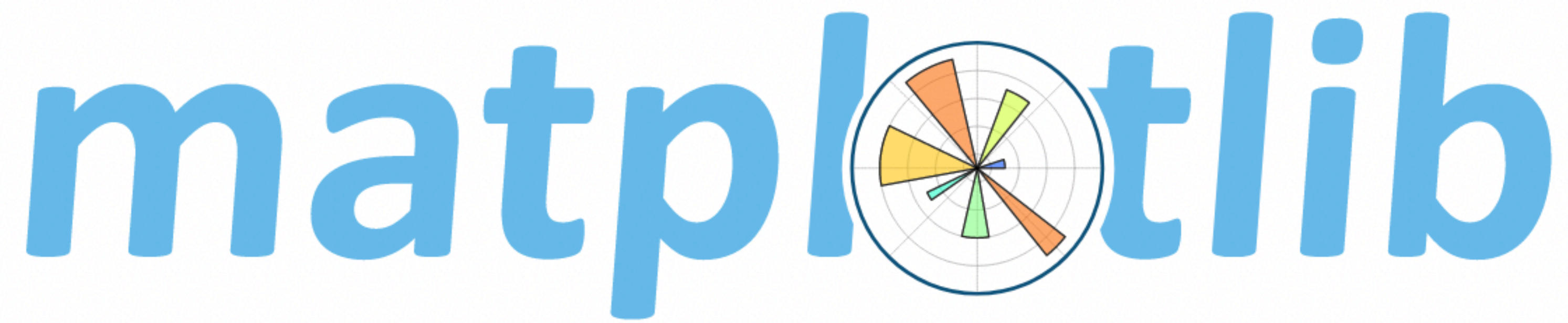




<https://matplotlib.org/stable>

Matplotlib

Installation

- Installation mittels pip:

```
pip install matplotlib
```

- Installation mittels conda:

```
conda install -c conda-forge matplotlib
```

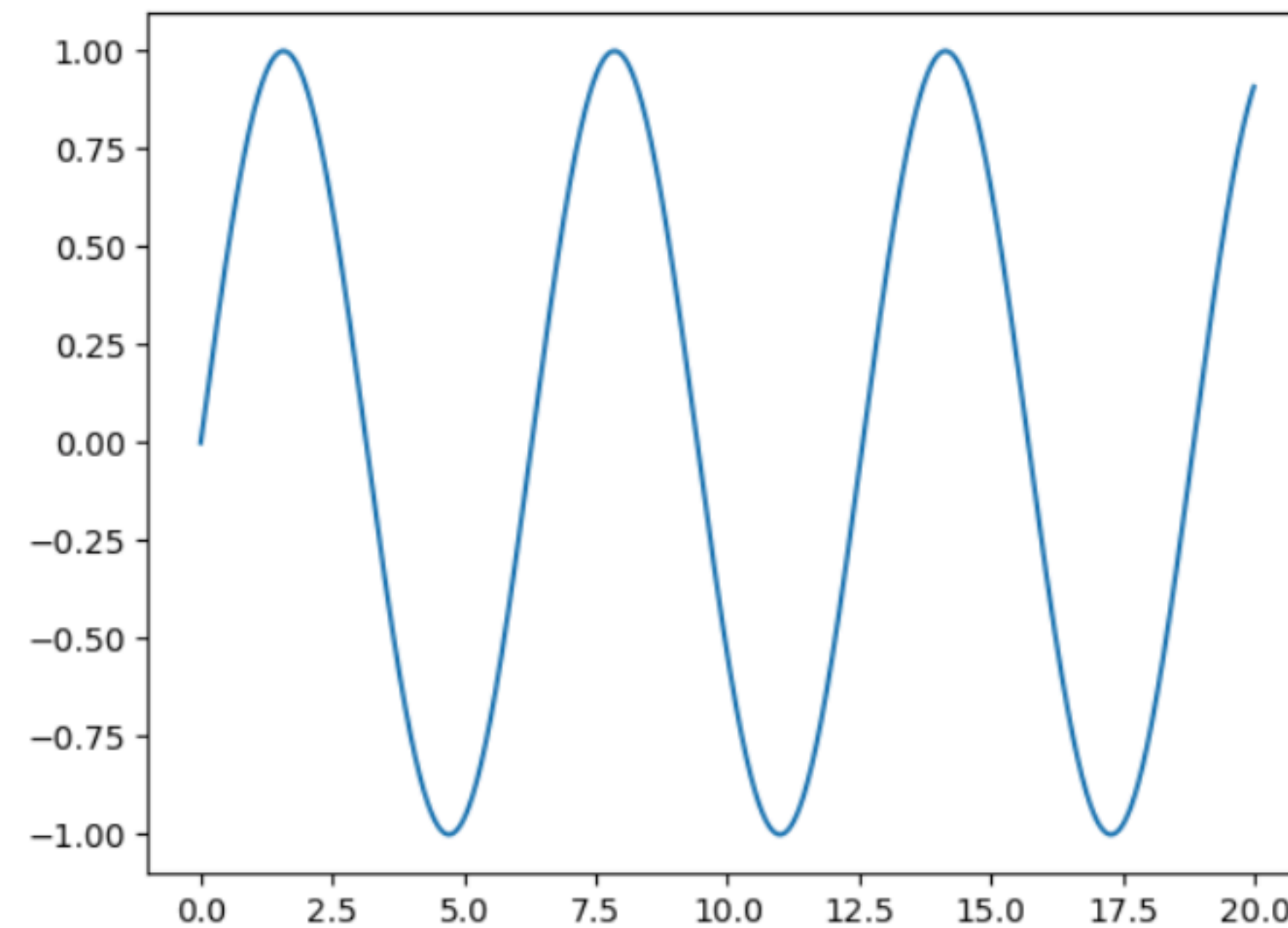
Matplotlib

Ein erstes Beispiel mit "plot" (geeignet z.B für Mathematische Funktionen)

```
import numpy as np
import matplotlib.pyplot as plt

# definition der (x,y) Wertpaare
x = np.arange(0,20, 0.01) #
y = np.sin(x)

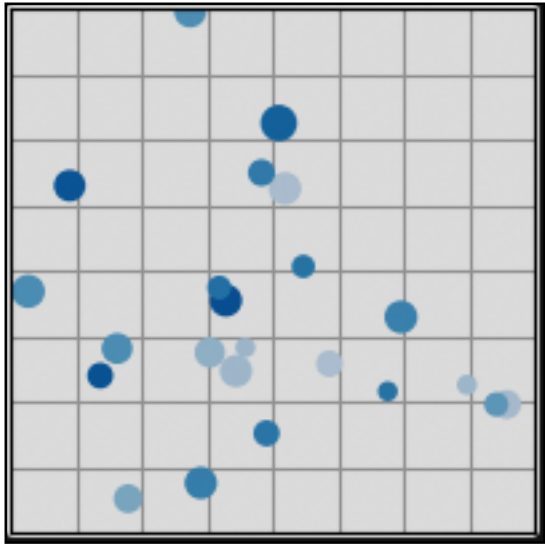
# schreiben der (x,y)-Wertpaare
plt.plot(x, y)
plt.show()
```



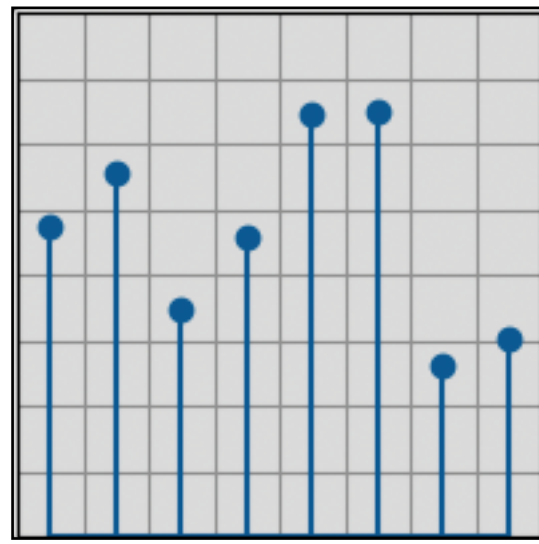
vgl.: matplotlib-examples.ipynb

Matplotlib

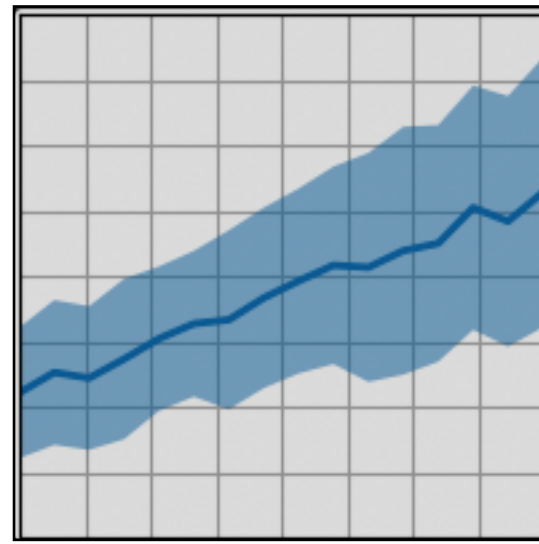
Neben "plot" gibt es noch eine Vielzahl anderer Plot Types



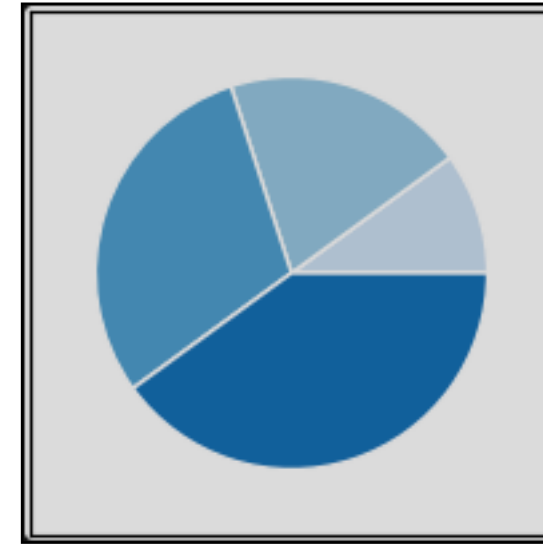
scatter(x, y)



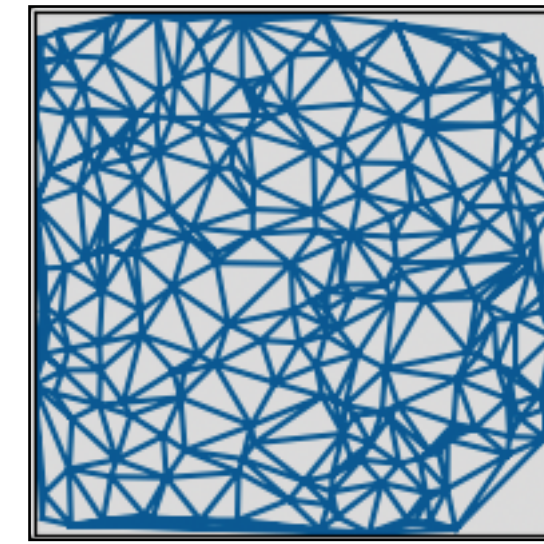
stem(x, y)



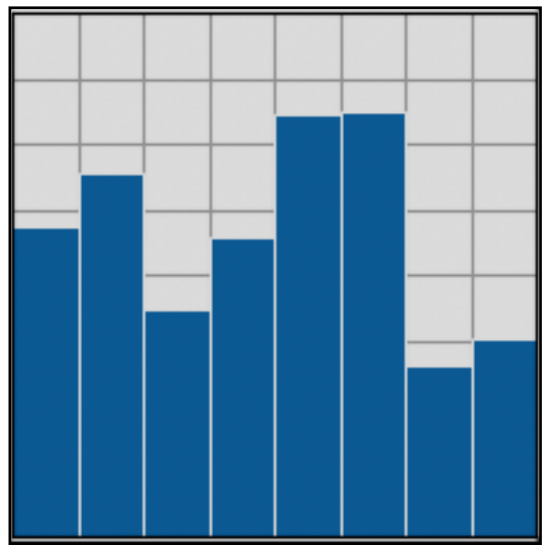
fill_between(x, y1, y2)



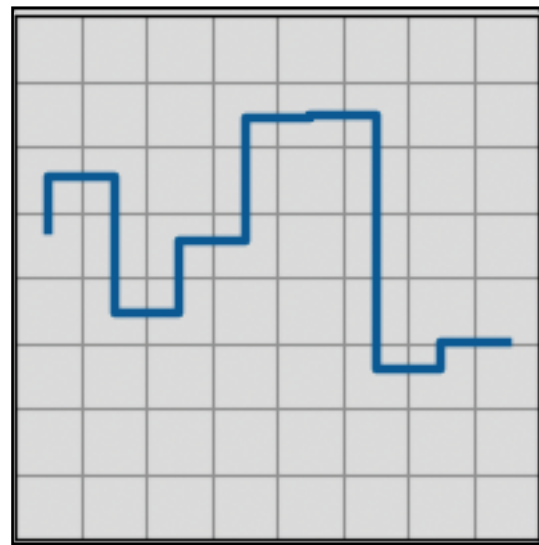
pie(x)



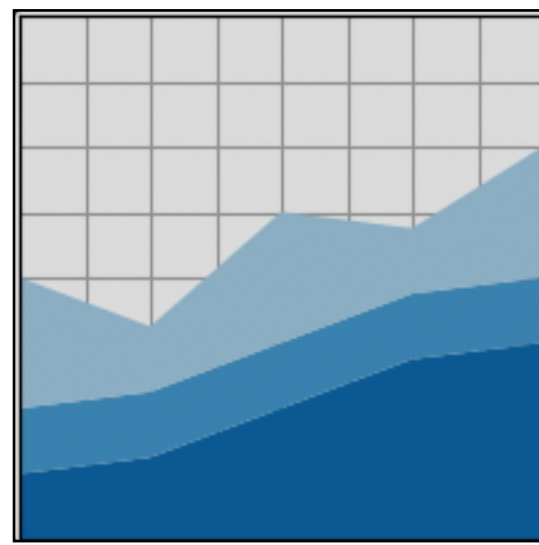
triplot(x, y)



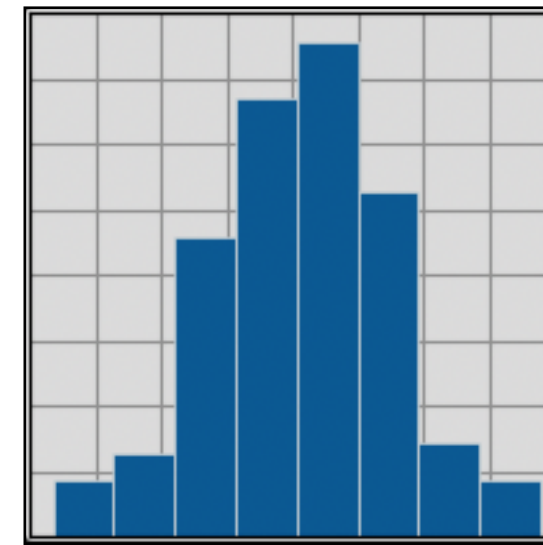
bar(x, height)



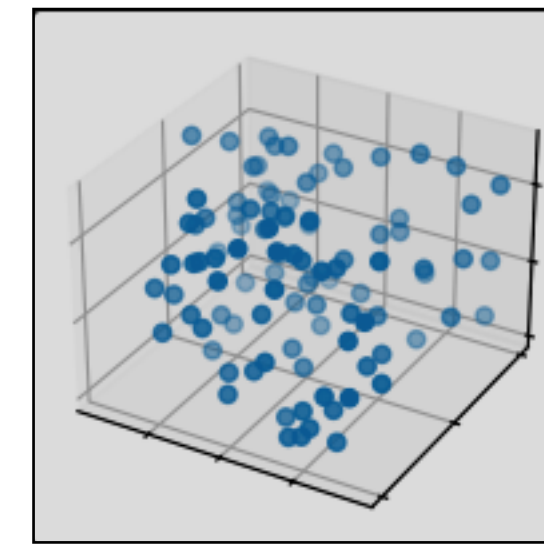
step(x, y)



stackplot(x, y)



hist(x, y)

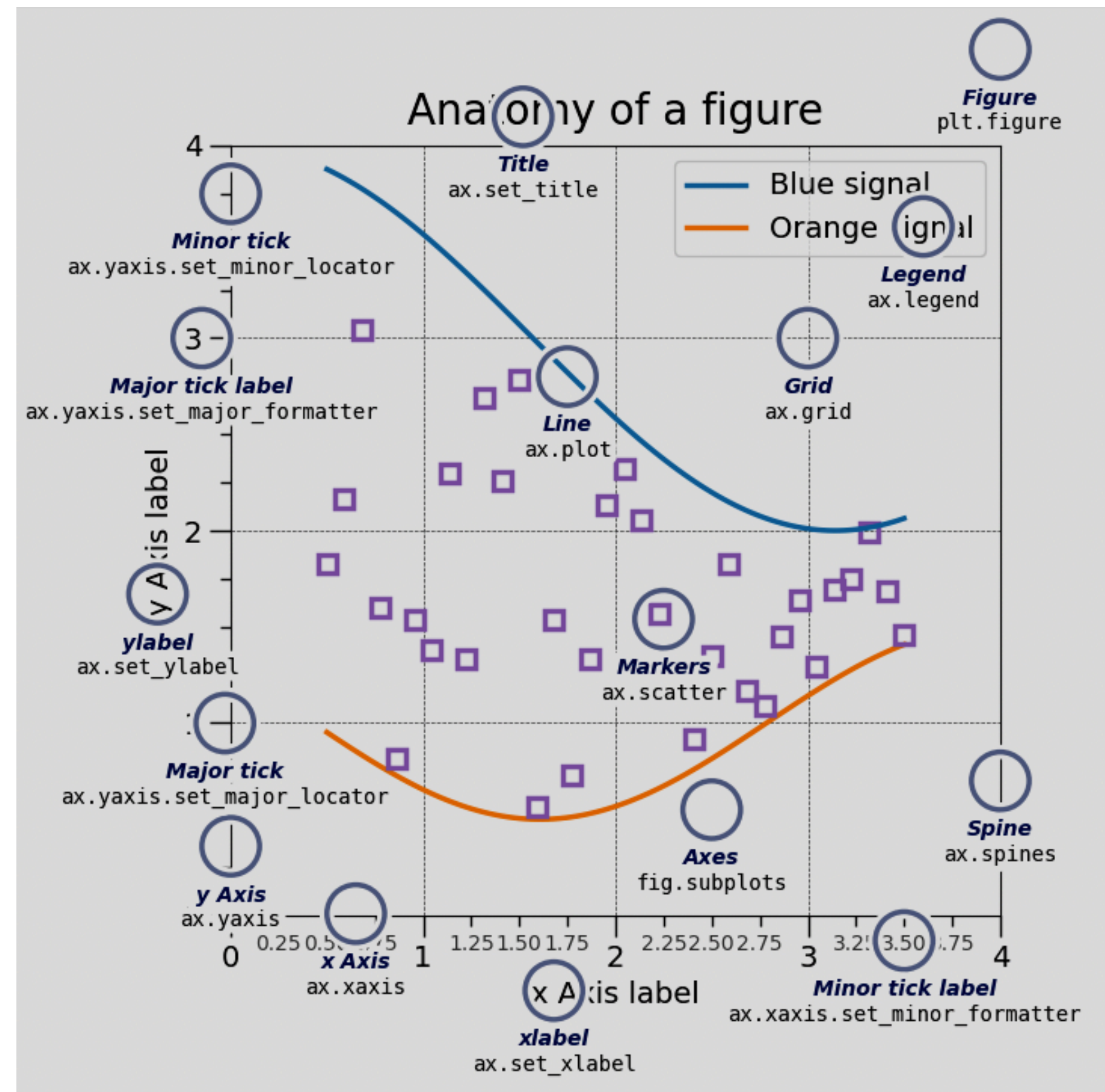


3D-scatterplot

vgl.: https://matplotlib.org/stable/plot_types/unstructured/index.html

Matplotlib

Figure Anatomy



vgl.: <https://matplotlib.org/stable/gallery/showcase/anatomy.html>